12TH CHEMISTRY · CHAPTER 33 · FULL CHAPTER NOTES

Water

Topic-wise short and long questions with answers from the complete chapter

33.1 PROPERTIES AND QUALITY OF WATER

Q.1 [Short] Why is water called the foundation of life?

Water is the foundation of life. It keeps humans alive and healthy through hydration and hygiene and removes toxic materials from the body; it allows plants to carry out photosynthesis and supports agriculture; animals use it to regulate body functions and microorganisms depend on it. In nature, water shapes the land, balances the climate and recharges underground reservoirs. It is also vital for energy production, transport, urban development and research. Water scarcity threatens the environment, communities and economies.

Q.2 [Short] Why are large bodies of water thermally stable?

The water molecule (H_2O) is a permanent dipole. This polarity leads to hydrogen bonds between water molecules. Hydrogen bonding gives water a high specific heat capacity and a high heat of vaporisation, so large bodies of water are thermally stable and resist rapid temperature changes.

Q.3 [Short] How does the solvent ability of water relate to water pollution?

Because of its polarity, water is a good solvent for a wide range of ionic substances such as metallic ions, nitrates (NO_3^-) and phosphates (PO_4^{3-}); it surrounds the ions. Water also dissolves polar organic compounds, so many pesticides and herbicides easily enter the water cycle, where they can persist and undergo bioaccumulation.

Q.4 [Long] Explain the amphoteric nature of water. How does acid rain affect water quality?

Water can act as both an acid and a base. Its self-ionisation ($K_w = 1.0 \times 10^{-14}$ at 25 °C) provides the baseline for the pH scale. Most aquatic life is highly sensitive to pH changes. Acid rain, caused by dissolved SO_2 and nitrogen oxides (NO_x), increases the concentration of hydronium ions (H_3O^+). This acidity mobilises toxic heavy-metal compounds such as aluminium hydroxide and lead salts from sediments into the water, by increasing their solubility through protonation or by displacing them from mineral surfaces.

$$2H_2O_{(l)} \rightleftharpoons H_3O^+_{(aq)} + OH^-_{(aq)}$$

Q.5 [Long] Explain DO, BOD and COD as indicators of water quality.

DO	BOD	COD
Dissolved oxygen: the amount of oxygen available in water for aquatic organisms to breathe.	Biochemical oxygen demand: the oxygen required by microorganisms to break down organic matter in water over a specific period, usually five days at 20 °C.	Chemical oxygen demand: the oxygen required to chemically oxidise organic and inorganic substances in water using oxidising agents ($KMnO_4$, $K_2Cr_2O_7$, etc.).
Key indicator of water quality and ecosystem health.	Key indicator of organic pollution.	Shows the total pollution load.
High DO = clean, unpolluted water; low DO = pollution or heavy decomposition of organic matter.	High BOD = much organic matter, heavy pollution; it depletes dissolved oxygen and harms aquatic life.	High COD = many oxidisable chemical pollutants; poor water quality, contamination by industrial or domestic waste.

Q.6 [Short] Give the typical ranges of DO, BOD and COD in different types of water.

Water type	DO (mg/L)	BOD (mg/L)	COD (mg/L)
Very pure water	7 – 10	< 1	< 2
Drinking water	> 6	< 2	< 5
Moderately polluted	4 – 6	2 – 8	10 – 20
Treated sewage	2 – 4	10 – 30	30 – 100
Raw sewage	< 1	100 – 400	200 – 1000
Industrial waste	0 – 2	500 – 5000+	1000 – 20,000+

Q.7 [Short] What is the redox potential of water? What happens in oxygen-depleted water?

The redox potential of water is a measure of the oxidising or reducing nature of a water body. In well-oxygenated water, chemical species exist in their most oxidised forms, such as SO_4^{2-} and CO_2. In polluted or stagnant water, where oxygen is depleted, the environment becomes reducing and toxic substances such as hydrogen sulphide (H_2S) and ammonia (NH_3) are formed.

Quick Check 33.1

Q.8 [Short] **a) Explain how the polarity of water facilitates the transport of agricultural pollutants like pesticides into the water cycle.**

Water molecules are permanent dipoles, so water dissolves ionic substances and polar organic compounds by surrounding them. Many pesticides and herbicides therefore dissolve in rain and irrigation water and are carried with it into rivers, lakes and groundwater, where they persist and bioaccumulate.

Q.9 [Short] **b) Differentiate between Biological Oxygen Demand (BOD) and Chemical Oxygen Demand (COD) as indicators of water quality.**

BOD	COD
Oxygen required by microorganisms to break down organic matter (usually 5 days at 20 °C).	Oxygen required to chemically oxidise organic and inorganic substances using oxidising agents ($KMnO_4$, $K_2Cr_2O_7$).
Indicates the level of organic (biodegradable) pollution.	Indicates the total load of oxidisable pollutants.

33.2 SOURCES OF WATER POLLUTION

Q.10 [Short] **What is point source pollution? Give examples.**

Point-source pollution is contamination that enters a water body from a single, discrete and identifiable location. Because the discharge is concentrated at one point, it often causes a **pollution plume**: at the site of discharge or immediately downstream the pollutant concentration is highest, causing localised acute toxicity. **Examples:** a factory discharging lead or mercury salts into a river through a wastewater pipe; a storage tank at a chemical plant leaking into the soil and then the groundwater.

Q.11 [Short] **What is non-point source pollution? Give examples.**

Non-point source (NPS) or diffused pollution occurs when rainfall or snowmelt moves over and through the ground, picks up natural and human-made pollutants and finally deposits them in lakes, rivers, wetlands and coastal waters. It depends heavily on weather events such as heavy storms and gives a mixture of pollutants at lower concentrations over a much wider area. **Examples:** rain washing excess ammonium nitrate (NH_4NO_3), fertilizers and pesticides from thousands of hectares of farmland into a river; stormwater moving over roads and car parks collecting leaked engine oils (hydrocarbons), zinc from tyres and road salts.

Q.12 [Short] **Differentiate between point and non-point sources of pollution.**

Point source	Non-point source
Enters from a single, identifiable location (pipe, tank).	Comes from many diffuse places over a wide area.
Pollutant is concentrated at the site of entry (pollution plume, acute toxicity).	A mixture of pollutants at lower concentrations.
Does not depend on the weather.	Depends on rainfall, snowmelt and storms.
Example: factory wastewater pipe.	Example: farmland and road run-off.

Quick Check 33.2

Q.13 [Short] **a) Classify these as point or non-point sources of water pollution, explain your choice.**
i) The outfall pipes used to release treated sewage into a bay or river from wastewater treatment plants.
ii) The "acid rain" caused by the nitrogen oxides (NO_x) from distant traffic or power stations dissolved in rainwater.

i) Point source: the sewage enters the water from a single, identifiable location (the pipe).

ii) Non-point source: the gases come from many distant, scattered sources and the acid falls with rain over a wide area; no single point of entry can be identified.

Q.14 [Short] **b) Give two major differences between point and non-point pollution sources.**

Point source	Non-point source
Enters from a single, identifiable location (pipe, tank).	Comes from many diffuse places over a wide area.
Pollutant is concentrated at the site of entry (pollution plume, acute toxicity).	A mixture of pollutants at lower concentrations.
Does not depend on the weather.	Depends on rainfall, snowmelt and storms.
Example: factory wastewater pipe.	Example: farmland and road run-off.

33.3 COMMON WATER POLLUTANTS

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Q.15 [Short] What are microbiological pollutants? Which diseases do they cause?

Microbiological pollution is the contamination of water by harmful microorganisms (bacteria, viruses, protozoa and parasites). It results from the discharge of untreated or poorly treated sewage, animal waste, hospital waste and agricultural run-off. Common pathogens are *Escherichia coli*, *Salmonella* and *Giardia*. They cause water-borne diseases such as cholera, typhoid, dysentery and hepatitis A. Boiling (100 °C) kills most bacteria, viruses and protozoa, but it cannot remove other pollutants.

Q.16 [Long] What are the sources of heavy metals in water?

Toxic heavy metals such as mercury, lead, arsenic and cadmium enter water through industrial discharge, mining operations and improper waste disposal.

Metal	Sources
Mercury	Coal combustion, gold mining, industrial waste from thermometer or battery manufacturing. In sediments, bacteria convert inorganic mercury into the more dangerous methylmercury, $[CH_3Hg]^+$.
Lead	Leaded petrol (now banned worldwide), old plumbing (lead pipes), lead-acid batteries and paints. Pb^{2+} ions often enter water through corrosion of pipes.
Cadmium	Metal plating, plastic stabilisers and impurities in phosphate fertilizers.

These pollutants are non-biodegradable and accumulate in the food chain, posing long-term risks to aquatic life and humans.

Q.17 [Long] Describe the health effects of heavy metals.

Metal	Effect on the human body
Mercury (as methylmercury)	Highly fat-soluble, so it crosses biological membranes and the blood–brain barrier and is biomagnified. A potent neurotoxin; long exposure causes **Minamata disease** (ataxia, numbness in hands and feet, muscle weakness).
Lead	Interferes with the production of haemoglobin, causing anaemia. In children it is a cumulative poison causing irreversible neurological damage and developmental delays.
Cadmium	Long-term ingestion causes **Itai-itai disease**: Cd^{2+} replaces Ca^{2+} in bones, making them extremely brittle. It also causes severe kidney damage (nephrotoxicity).

Q.18 [Short] Why can cadmium ions replace calcium ions in bones?

Cd^{2+} and Ca^{2+} have the same charge and similar size, and the book states that Cd^{2+} ions have a higher charge density, so they can displace Ca^{2+} ions from bone (ionic displacement). The bones become extremely brittle (Itai-itai disease).

Q.19 [Short] What are the WHO permissible limits of heavy metals in drinking water?

Heavy metal	WHO limit (mg/L)
Arsenic (As)	0.01
Chromium (Cr)	0.05
Copper (Cu)	2.0
Lead (Pb)	0.01
Cadmium (Cd)	0.003
Mercury (Hg)	0.006

Q.20 [Short] What was the Minamata disaster?

The Minamata disaster was a severe case of industrial pollution in the 1950s in Minamata, Japan. Methylmercury released into the water caused widespread mercury poisoning; thousands of people suffered neurological damage, and there were birth defects, deaths and environmental devastation.

Q.21 [Short] What are the sources and nature of pesticide pollution in water?

Pesticides and herbicides such as DDT and glyphosate come mainly from agricultural run-off and domestic gardening. They are non-polar and insoluble in water but highly soluble in the fatty tissues of fish and birds. Many older pesticides like DDT are **persistent organic pollutants (POPs)**. Polychlorinated biphenyls (PCBs), used as coolants in electrical transformers, leak into water from hazardous waste sites.

The textbook prints "Polychlorinated biphenols"; the standard name of PCBs is polychlorinated biphenyls. It also says pesticides "dissolve as polar organic compounds" in 33.1 but calls them non-polar here; both statements are kept as printed in their own sections.

Q.22 [Short] What are the health effects of pesticides and PCBs?

- Many pesticides act as **endocrine disruptors**, behaving like natural hormones such as estrogen; they are linked to hormonal disruption, developmental problems and hormonal cancers in humans.
- In wildlife they cause reproductive failure, e.g. thinning of eggshells in birds.
- PCBs are probable human carcinogens; they also suppress the immune system and cause skin conditions such as chloracne.
- Chemical pollution also degrades water quality, reduces biodiversity and disrupts aquatic ecosystems.

Q.23 [Long] What is eutrophication? Explain its causes and effects.

Eutrophication (nutrient pollution) occurs when excessive nutrients, mainly nitrogen and phosphorus, enter water bodies from agricultural run-off, wastewater discharge and stormwater.

1. Excess nutrients cause rapid growth of algae (algal blooms).

2. The blooms block sunlight from reaching underwater plants.
3. When the algae die and decompose, large amounts of oxygen are consumed.
4. Hypoxic “dead zones” form, where aquatic life cannot survive.

It disrupts ecosystems, reduces biodiversity, harms fisheries and spoils water for recreation and drinking. High nitrate levels in drinking water cause **Blue Baby Syndrome** (methaemoglobinaemia) in infants: nitrates reduce the blood’s capacity to carry oxygen.

Q.24 [Short] How does oil pollute the oceans and harm marine life?

Oil from leaks and spills of ships spreads rapidly over the sea surface, forming a layer that blocks sunlight and reduces oxygen exchange. This severely harms fish, birds and mammals: marine mammals ingest the toxic oil while grooming, and fish suffer long-term health problems from the hydrocarbons. Oil also damages coastal habitats such as mangroves, coral reefs and beaches.

Q.25 [Short] Why is plastic and solid waste a growing environmental crisis?

Millions of tons of plastic waste (bottles, bags, fishing nets, microplastics) are dumped into rivers, lakes and oceans every year. Unlike organic materials, plastics do not biodegrade; they break into smaller particles that persist for centuries. Marine animals mistake plastic for food, leading to ingestion, choking or internal injury; entanglement in old fishing gear injures or kills turtles, dolphins and birds. Microplastics have entered the food chain, raising concerns for human health. Solid waste also clogs waterways, disrupts habitats and reduces water quality.

Quick Check 33.3

Q.26 [Short] i. Name the sources of heavy metals.

Industrial discharge, mining and improper waste disposal. Mercury: coal combustion, gold mining, thermometer and battery manufacturing. Lead: old lead pipes, lead-acid batteries, paints (and formerly leaded petrol). Cadmium: metal plating, plastic stabilisers and phosphate fertilizer impurities.

Q.27 [Short] ii. How oil spill is hazardous to the marine life?

Oil from leaks and spills of ships spreads rapidly over the sea surface, forming a layer that blocks sunlight and reduces oxygen exchange. This severely harms fish, birds and mammals: marine mammals ingest the toxic oil while grooming, and fish suffer long-term health problems from the hydrocarbons. Oil also damages coastal habitats such as mangroves, coral reefs and beaches.

Q.28 [Short] iii. Pesticides like DDT are “non-polar and insoluble in water” yet they “pollute water and cause severe human health issues”. Explain.

Although DDT does not dissolve in water, it is carried into water bodies by agricultural run-off and it persists there (it is a persistent organic pollutant). Being non-polar, it dissolves in the fatty tissues of fish and birds, accumulates and is magnified up the food chain. Humans who eat these organisms receive high doses, which cause hormonal disruption and are linked to cancers.

33.4 POLLUTION DUE TO HUMAN ACTIVITIES

Q.29 [Long] **Evaluate the impact of human activities on water resources.**

Activity	How it pollutes water
Industrial	Factories (oil refineries, chemical plants, automobile industry, fertilizers, tanneries) discharge harmful chemicals, heavy metals and toxic waste directly into rivers, lakes and oceans.
Agricultural	Run-off of fertilizers, pesticides and herbicides; poor irrigation increases soil erosion, washing sediments into rivers; farm drainage contaminates surface and groundwater.
Domestic and municipal waste	Untreated or poorly treated sewage and solid waste bring harmful bacteria and viruses; household detergents, oils and chemicals enter drains; plastics block water channels.
Deforestation and urban development	Removing trees loosens soil; erosion adds sediment that clogs waterways and harms habitats; construction adds pollutants and debris to run-off.
Recreational	Boats release oil, fuel and chemicals; visitors litter plastics, food wrappers and other waste.
Marine transport	Discharge of oil, fuel and chemicals from ships; accidental oil spills; wastewater and sewage from vessels; hull cleaning and painting release toxic substances.

Quick Check 33.4

Q.30 [Short] **a) A river passes by three regions in a country: a rural area with vast agricultural land and a lot of cattle; an industrial area with many paint and battery industries; and a large residential area of a big city:**
i) Predict which type of pollutants will be added to the river in the rural area; organic, inorganic, or both? Name two of them.
ii) Which metals are most likely to pollute the water near the industrial area?
iii) Name the regions that will contaminate the water with chemicals that lead to Eutrophication.
iv) Name four contaminants that will pollute the water near the residential area.

i) Both. Organic: pesticides and animal (cattle) waste. Inorganic: fertilizers such as ammonium nitrate and phosphates.

ii) Lead (paints, lead-acid batteries), together with mercury and cadmium (battery manufacturing).

iii) The rural area (nitrogen and phosphorus from fertilizers and animal waste) and the residential area (sewage and detergents).

iv) Sewage (with bacteria and viruses), detergents, oils and household chemicals, and plastics/solid waste.

33.5 EFFECT ON THE ENVIRONMENT

Q.31 [Long] Explain the effects of water pollution on the environment.

Pollutants alter the physical and chemical properties of water, making it difficult for plants and animals to survive.

1. **Destruction of ecosystems and biodiversity:** oil, heavy metals and pesticides reduce oxygen levels (hypoxia), damage coral reefs and disrupt food chains, breeding cycles and migration; populations decline and sensitive species may become extinct.
2. **Nutrient overload (eutrophication):** excess nitrogen and phosphorus from fertilizers and sewage cause rapid algal growth, which blocks sunlight, lowers oxygen and kills fish.
3. **Bioaccumulation and biomagnification:** toxins such as mercury, lead and pesticides build up in organisms and become more concentrated up the food chain; top predators, including humans, receive the highest doses.
4. **Soil degradation and plant damage:** heavy metals, industrial waste and excess nutrients seep into the soil, change its structure and fertility and disturb its microbes; plants show stunted growth, leaf discoloration and may die.
5. **Methane emission:** organisms fed by sewage and run-off die and decompose in oxygen-poor (anaerobic) conditions, releasing methane, a potent greenhouse gas that worsens global warming.

Q.32 [Short] Differentiate between bioaccumulation and biomagnification.

Bioaccumulation	Biomagnification
Build-up of a toxic substance in the body of one organism over time.	Increase in the concentration of the toxin at each step up the food chain.
Occurs within an organism.	Occurs between trophic levels; top predators, including humans, receive the highest doses.

Quick Check 33.5

Q.33 [Short] **a) Methylmercury ($[CH_3Hg]^+$) is "highly fat-soluble," allowing it to cross biological membranes and the blood-brain barrier.**
i. Contrast the chemical structure of the methylmercury cation ($[CH_3Hg]^+$) with the inorganic mercury ion (Hg^{2+}). Explain why the presence of the methyl group ($-CH_3$) alters its solubility in lipid bilayers of living cells compared to the inorganic ion.
ii. Predict the impact on the "biomagnification" process if ($[CH_3Hg]^+$) was highly water-soluble but had low fat solubility.

i. Hg^{2+} is a simple ion with a 2+ charge; it is strongly hydrated by polar water molecules and cannot dissolve in non-polar lipids. In $[CH_3Hg]^+$ a non-polar methyl group is covalently bonded to mercury and the charge is only 1+. The non-polar $–CH_3$ group gives the ion an organic (lipophilic) character, so it dissolves in the non-polar lipid bilayer and passes through membranes.

ii. It would stay dissolved in body water and be excreted (e.g. in urine) instead of being stored in fatty tissue. It would not accumulate in organisms, so its concentration would not increase up the food chain; biomagnification would be very small.

The explanation uses the "like dissolves like" principle; the chapter states only that methylmercury is highly fat-soluble.

Q.34 [Short] **b) How the water pollution can contribute to the global warming?**

Sewage, agricultural run-off and industrial waste boost the growth of algae and other organisms. When these die and decompose in oxygen-poor (anaerobic) conditions, methane is released. Methane is a potent greenhouse gas, so it adds to climate change and global warming.

33.6 WASTEWATER TREATMENT

Q.35 [Short] **What are the three levels of wastewater treatment?**

Different methods are used according to the type and severity of pollution (organic matter, chemicals or pathogens). They are applied at three levels: **primary treatment** (removal of physical debris), **secondary treatment** (biological processing) and **tertiary treatment** (chemical purification).

Q.36 [Long] **Describe the methods and technologies used in water treatment.**

i. Screening and sedimentation: large insoluble particles are removed by passing water through metal screens. The water is then held in large tanks so that suspended solids settle at the bottom under gravity.

ii. Coagulation and flocculation: used, if required, to speed up the settling of finer colloidal particles and to precipitate dissolved contaminants (heavy metals, phosphates). A chemical coagulant such as aluminium sulphate, $Al_2(SO_4)_3$, makes the small particles clump together into larger masses called "floc", which are easily removed.

iii. Filtration: a mechanical process that removes the remaining fine suspended matter and some microorganisms. Water passes through layers of graded sand and gravel; the sand traps particles in the pores between grains. Activated carbon is used for more advanced purification.

iv. Disinfection: filtration cannot remove dissolved ions or microscopic pathogens, so chemical purification is needed. Chlorination is the most common method; sodium hypochlorite (NaOCl), ozone (O_3) or hydrogen peroxide (H_2O_2) are also used. Chlorine or sodium hypochlorite forms hypochlorous acid (HOCl), which penetrates the cell walls of bacteria and viruses and disrupts their functions.

$$Cl_{2(g)} + H_2O_{(l)} \rightleftharpoons HOCl_{(aq)} + HCl_{(aq)}$$

v. Reverse osmosis (RO) and ion exchange: used for high-purity water (laboratories, regions of high salinity). In RO, water is forced under pressure through a semi-permeable membrane that filters out dissolved salts, bacteria and other impurities. Ion exchange "softens" hard water: the Ca^{2+} and Mg^{2+} ions are swapped for Na^+ or H^+ ions in an ion-exchange column.

The textbook numbers both Disinfection and Reverse Osmosis as "iv". Figure 33.4 shows the plant sequence: coagulation → flocculation → sedimentation (sludge removed) → filtration → chlorination and disinfection → treated water.

Q.37 [Short] Differentiate between reverse osmosis and ion exchange.

Reverse osmosis (RO)	Ion exchange
Water is forced under pressure through a semi-permeable membrane.	Water passes through an ion-exchange column where ions are swapped.
A physical (filtering) process.	A chemical exchange of ions.
Targets dissolved salts, bacteria and other impurities.	Targets hardness ions Ca^{2+} and Mg^{2+}, replaced by Na^+ or H^+.

33.7 PROTECTION AND CONSERVATION OF WATER

Q.38 [Short] Which strategies are used for pollution control?

- Imposing strict discharge standards on industries and municipalities.
- Upgrading and expanding sewage treatment plants (STPs), installing effluent treatment plants (ETPs) and adopting modern technologies such as membrane filtration and bioreactors.
- Continuous water-quality monitoring, so that pollutants are detected and contamination is answered in time.
- Strict penalties for polluters and incentives for clean practices.

Q.39 [Long] Explain the strategies for efficient use and conservation of water.

- Water-saving technologies in agriculture: drip irrigation and sprinkler irrigation.

- Reusing and recycling greywater (laundry, dishwashing and bathing water) for landscaping, toilet flushing and industrial cooling.
- Rainwater harvesting systems in urban and rural areas.
- Regular repair and maintenance of water-supply infrastructure to minimise leaks.
- Constructing small dams and recharge wells to store rainwater and replenish groundwater.
- Community engagement and awareness through public education campaigns.

33.8 RULES AND REGULATIONS

Q.40 [Long] Explain the laws and regulations related to water pollution and conservation.

The Government of Punjab has made several acts, and the Environmental Protection Agency (EPA) Punjab issues the related rules.

Act / Regulation	Primary scope and powers	Penalties
Punjab Environmental Protection Act, 1997	Pollution prevention.	Fines up to Rs 5 million, plus additional daily fines of up to Rs 100,000.
Punjab Water Act, 2019	Prohibits discharging pollutants or blocking flow.	Up to 3 months imprisonment and/or a Rs 500,000 fine.
Punjab Irrigation, Drainage and Rivers Act, 2023	Sets water usage and drainage fees; regulates discharge permits to prevent industrial/agricultural pollution.	—
Punjab Water and Sanitation Authority Act, 2025	Establishes the Authority in major cities (Lahore, Multan, etc.) for integrated resource management and groundwater conservation.	—

The penalty for using hosepipes for car-washing and flood irrigation of domestic lawns or gardens is a fine of Rs 10,000 (as stated in the textbook, under the Pakistan Penal Code).